

Joseph Barbosa

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EDUCATION

University of Waterloo & Wilfrid Laurier University

Bachelor of Computer Science (UW), Bachelor of Business Administration (WLU)

GPA: 3.98 / 4.0 (92%)

Sep. 2025 – Present

EXPERIENCE

Software Engineering Intern

Scotiabank

May 2026 – Aug. 2026

Toronto, ON

- Engineer on the production consumer banking platform (React, Spring); actively seeking Agentic AI adjacent initiatives

Software Engineering Intern

ESGTree

Jan. 2026 – Apr. 2026

Waterloo, ON

- Built a production NL-to-SQL agent over a **98-table** MySQL DB using a custom ReAct loop on the Anthropic Messages API with FAISS schema retrieval and FK-aware join hints; designed an eval suite of **77** user queries from production logs achieving **90.9%** accuracy, with ablations showing self-correction and retrieval contributed **+13.0pp** and **+7.8pp** respectively
- Engineered backend with read-only DB privileges, regex SQL validation, and prompt-level constraints; enforced multi-tenant isolation via JWT with engine-level separation of read (MySQL) and write (SQLite audit) paths; reduced post-query latency by delegating narrative summarization and chart spec generation to Haiku, preserving Sonnet budget for multi-step reasoning

Technical Project Manager

LuminWell

Sep. 2025 – Present

Waterloo, ON

- Led an **8+** member team of graduate students and undergraduate interns to build and deploy a Gemini-powered peer-matching platform (MERN stack), securing **\$2.5K** in external funding for mental health access initiatives

Full Stack Software Developer

University of Waterloo Computer Science Club

Sep. 2025 – Present

Waterloo, ON

- Maintained a production Next.js + TypeScript web application, implementing content updates across a shared codebase
- Collaborated with **8 teammates** using Git workflows (issues, pull requests) in a CI/CD-enabled development environment
- Engineered an LLM-powered Akinator game using Groq API, supporting **100+ concurrent users** with **<500ms latency**

Software Development Intern

FuturIQ

Jun. 2024 – Dec. 2024

Brampton, ON

- Built a production ML-powered real estate platform integrating a full-stack web application (Flask, PostgreSQL, JavaScript) with an automated price prediction pipeline that scraped **1000+** live housing market data points via Selenium, trained models on dynamically collected market subsets, and served predictions through the application backend
- Implemented an ML training and evaluation pipeline using train-test split, cross-validation, and RMSE, comparing linear and polynomial regression models and deploying a regularized polynomial model to improve generalization

PROJECTS

Scholar – AI Research Assistant

Python, OpenAI API, Pydantic, asyncio, OpenAlex API, MCP

2026

- Built a fully asynchronous LLM research pipeline over **200M+ papers** (OpenAlex), generating citation-grounded outputs
- Designed a multi-stage pipeline (decomposition → retrieval → synthesis) via Pydantic schema-validated LLM outputs
- Implemented recursive retrieval with query expansion and deduplication to improve coverage of mechanisms and limitations
- Improved user experience through graceful failure and query replanning; exposed as a CLI + MCP tool with **55 unit tests**

Studeal – Agentic Deal Discovery System

Python, FastAPI, LangGraph, Next.js, PostgreSQL + pgvector, Redis, Celery

2025 – Present

- Designed a multi-stage, LangGraph-based agentic retrieval system for deal discovery (retrieval, extraction, scoring)
- Built an asynchronous pipeline, parallelizing search, scraping, and inference, including semantic deduplication (pgvector)
- Implemented a scoring agent with market context for daily personalized deal matching from natural language watchlists
- Achieved **90%+ link resolution accuracy** by reconstructing missing product URLs using LLM-guided extraction
- Shipped production system with rate-limited APIs, Celery scheduling, and Stripe access control, plus Next.js UI

PantryPal – Fine-Tuned LLM for Constraint-Aware Generation

Python, FastAPI, Llama 3.2, QLoRA, DPO, XGBoost

2025 – Present

- Fine-tuned Llama 3.2 (3B) with QLoRA on **10K instruction samples** for constraint-aware recipe generation
- Built the synthetic dataset using **18 XGBoost classifiers** over ingredient features, enabling DPO without human labels

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, SQLite), TypeScript, HTML/CSS, Java, C

AI / ML: LangChain, LangGraph, Hugging Face, XGBoost, OpenAI/Groq APIs, RAG, pgvector, ChromaDB

Frameworks: React, FastAPI, Flask, Next.js, Node.js, Celery

Libraries: pandas, NumPy, scikit-learn, Pydantic, SQLAlchemy, Selenium, BeautifulSoup, Matplotlib

Tools: Git, Docker, Redis, Bash, VS Code, Figma, Postman, Supabase, Heroku, Railway

Certifications: Stanford Machine Learning Specialization (Supervised, Unsupervised, Reinforcement Learning)

Awards: President's Gold Scholarship, Loblaw National Scholarship, National Meritorious Award (IMMC), \$3,000+ in Case Competition Winnings (5+ finalist placements)